

DIMENSION

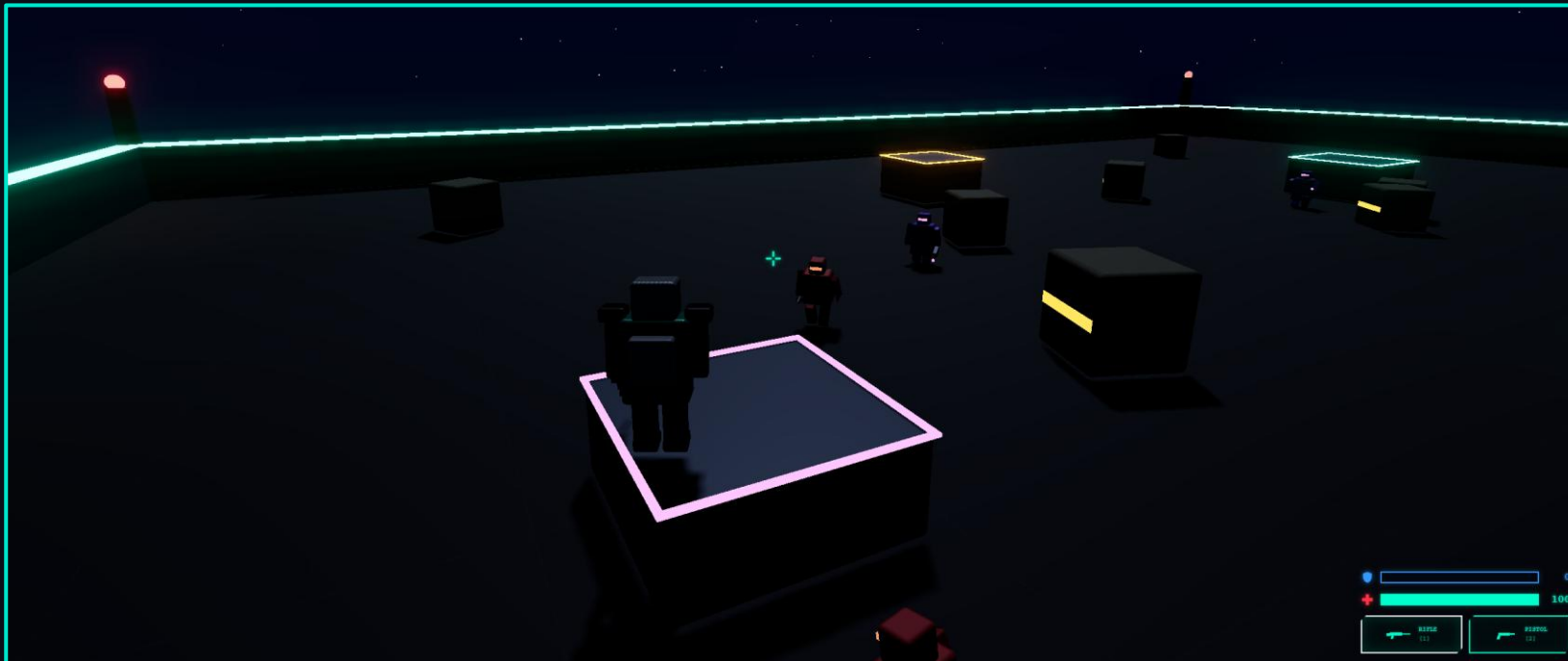
S H I F T

A Sci-Fi Arena Shooter in Three.js

MATTIA COSIMI — 2278125 — INTERACTIVE GRAPHICS

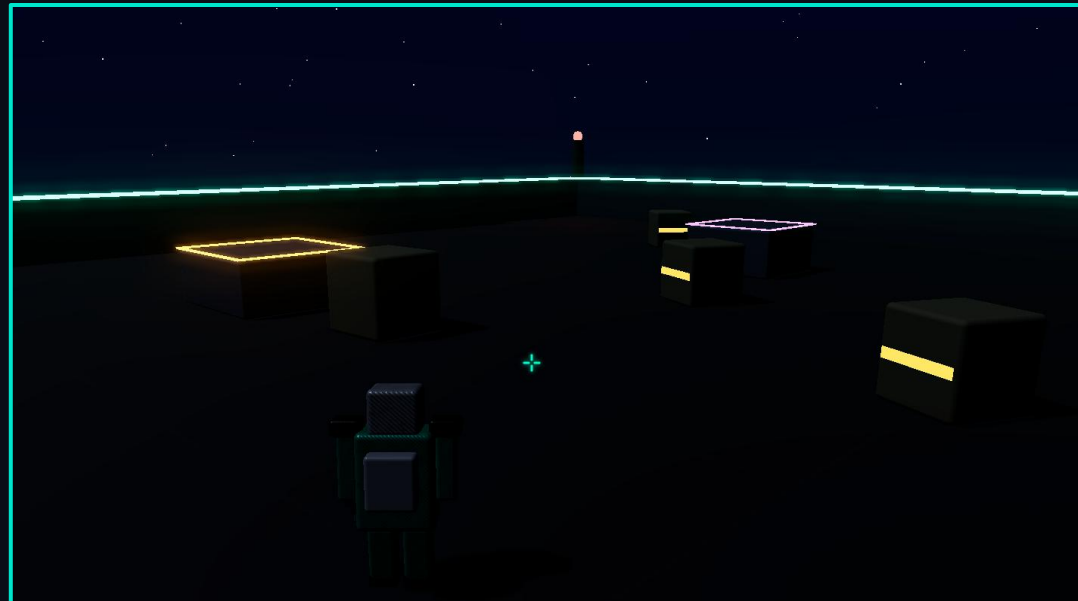
What This Actually Is

- Third person arena shooter — 9 waves + one boss fight
- 2 classes, 5 enemy types, one shared engine underneath
- One mechanic runs through everything I'll show you — I'll come back to it



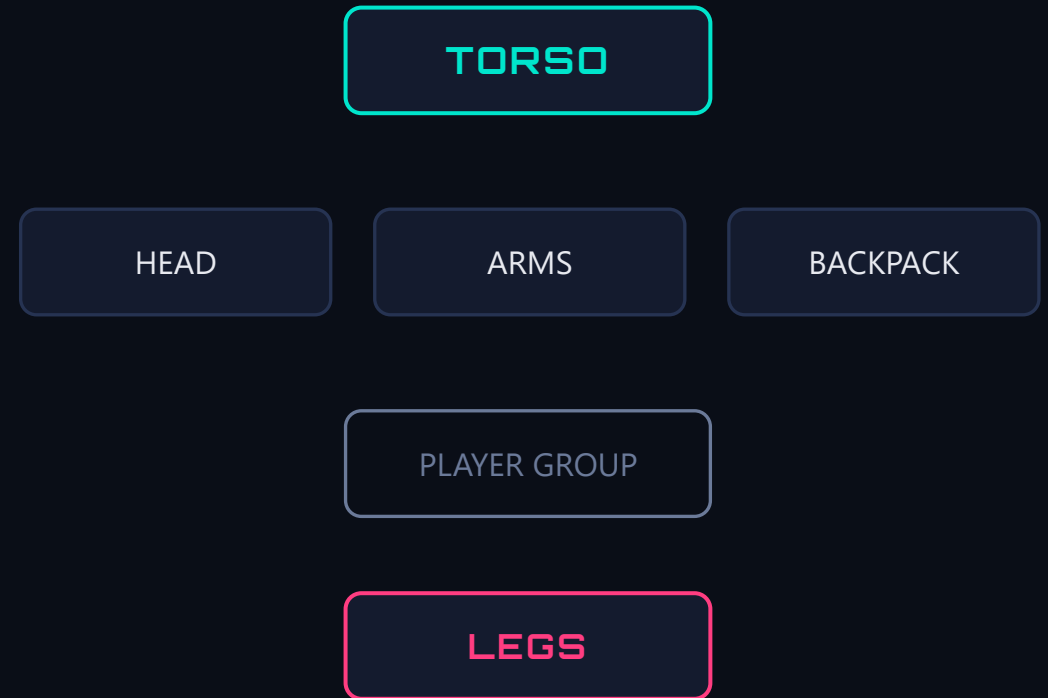
Building the Battlefield

- Procedural geometry only — no imported meshes anywhere
- One shared obstacle list, reused by every system
- Real jump physics: velocity solved from a target height
- 8 lights total, incl. one shadow-casting sun + a camera-glued fill light
- Bloom + filmic tone mapping sell the sci-fi glow



A Body Made of Boxes

- Torso is the hierarchy's root — every other part is its child
- Legs are children of the group, not the torso — they can stay planted while it twists
- Visor, chest core, glowing pauldrons, a backpack — small details, not just boxes



+ visor, chest core, pauldrons, backpack — detail geometry, not extra hierarchy

No Animation Files

- Every motion is a formula
- A sine wave drives each limb's swing, legs and arms in antiphase
- Evaluated fresh, every rendered frame — no stored poses at all



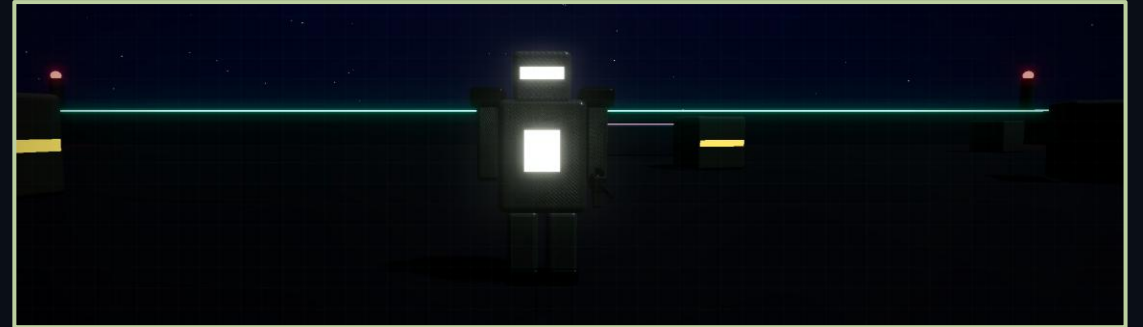
Choose Your Loadout

- Assault — Rifle + Pistol, teal armor tint
- Sniper — Sniper Rifle + Pistol, pale sage green tint
- One shared PlayerClass base — the game loop never branches on "which class", it just asks

ASSAULT



SNIPER



Three Guns, One Base Class

Every weapon extends one Weapon base — only these numbers and the 3D model change.

WEAPON	FIRE MODE	FIRE RATE	BULLET SPEED	DAMAGE
Rifle	Automatic	0.2 s	60 u/s	10
Sniper Rifle	Automatic	1.1 s	130 u/s	45
Pistol	Semi-auto	0.35 s	75 u/s	7

Rifle trades per shot damage for fire rate; Sniper Rifle inverts that trade entirely.

An Invisible Sphere Around You

- Over the shoulder orbit camera — two angles + one fixed distance
- Always looks back at the same fixed point on the player
- A screen space offset tucks the character toward the lower left



Where the Crosshair Really Points

- The gun sits at the hands — not where the camera is
- Every shot aims toward a point, not along a fixed angle
- Each frame: find what's under the crosshair, fire the gun at it



Staying Inside the World

- Every obstacle is one shape: a circle with a radius
- Movement is tested one axis at a time — X blocked, Z still slides
- A second pass every frame pushes out any leftover overlap



Five Threats, One Shared Brain

All five reuse the player's own movement, obstacle avoidance, and aiming code.

ENEMY	HP	SPEED	DAMAGE	RANGE	ROLE
Grunt	30	5.5	8	1.3	Fast, fragile melee rusher
Shooter	45	3	6/shot	13	Ranged, holds its distance
Brute	90	3.5	18	1.6	Heavy melee tank
Marksman	35	3.2	16/shot	18	Fragile kiter — retreats if you close in
Doomhorn	1500	4.2	26/bolt	15	Boss — twin-handed ranged bolts

Stats scale +10% per wave past the first — the same five types, gradually tougher.

Ten Waves, One Doomhorn

- Escalating difficulty, wave by wave
- Enemy mix shifts over time, not just enemy count
- Wave 10 is different: no crowd — just Doomhorn, alone



Resources, Not Just Pickups

One shared weighted random pool decides which pickup spawns next.

PICKUP	EFFECT	SPAWN WEIGHT
Health	Heals to full	4
Armor +25	Adds a flat 25 armor (capped at 100)	4
Armor (Full)	Refills armor to its maximum outright	1.5
Strength	+60% damage, +30% move speed, for 15 s	2
Dimension Cache	Full heal + full armor — invisible outside Toon mode	1.5

Armor absorbs damage before health does — a second pool, not a shield timer.

One Button. A Different Game.

- Realistic ↔ Toon, on TAB
- A real rendering swap, not a filter
- Materials, lights, sky, post processing — all repainted together

REALISTIC



TOON



A Resource, Not a Costume

- 15 s active, then a 25 s cooldown
- Enemies take bonus damage while it's active
- It's also the only way to see the Dimension Cache pickup

LET'S PLAY

